

RESEARCH ARTICLE

PruferNET: An Augmented Graph Neural Network for Influential Node Detection in Social Networks

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ABSTRACT Finding influential nodes, or people, on a social network is very important because it supports many important uses in different fields. This paper proposes a new way to classify social media nodes using Prufer sequence-based structural features with *Graph Neural Networks (GNNs)*. Using the Facebook dataset, a *Minimum Spanning Tree (MST)* is built to generate the Prufer sequence and extract graph-based features. To build a detailed feature set, properties such as label frequency, sequence entropy, subtree depth, unique label ratio, and transition influence are combined with traditional structural measures such as degree centrality, closeness centrality, and PageRank. The method employs a GraphSAGE model for node classification, where nodes are tagged as Very Shallow, Shallow, or Deep according to their relative depth in the reconstructed Prufer tree. The results show that the integration of Prufer-based features significantly improves performance. The method achieved a test precision of 99.63% after adding Prufer features with traditional feature vectors. This is considerably higher than the accuracy 96.54% obtained using only traditional features. The precision, recall, and F1 scores across all categories show the strength and reliability of the proposed method. The results highlight the effectiveness of this combined graph-based representation. The findings demonstrate improved performance, especially for those derived from spanning tree encodings such as Prufer sequences, with deep learning models. This system effectively captures hidden structural patterns and identifies influential nodes in large social graphs.

INDEX TERMS Prufer sequence, graph neural network (GNN), GraphSAGE, node classification, structural features, Facebook dataset.

I. INTRODUCTION

With the expansion of online social networks, there has been significant interest in finding the most important users and understanding how they are connected to others [1], [2], [3], [4]. These influential users often serve as a major factor in the spread of information or in the rapid connection of different groups. Identifying such individuals within a network facilitates various tasks such as targeted advertising, spreading news or ideas, and finding communities within the network [5], [6]. In recent years, online social media platforms (OSM) have become extremely popular and have grown rapidly. Today, millions of users express their thoughts and ideas and react to, comment on, share, or tag others in

posts on OSM platforms. Due to this high level of activity, data-driven organizations have increasingly turned their attention to social networks. By examining what individuals post, the frequency of their posts, and the types of content they interact with, organizations can derive valuable insights into consumer interests, preferences, and decision-making factors [7].

Identifying key individuals within a social network is crucial, especially when the objective is to promote the adoption of a new product that others may follow. Social networks can be characterized as static (unchanging) or dynamic (constantly evolving). Identifying influential people/nodes in static networks is comparatively easy, and there are several algorithms for this task. Recently, the focus has shifted toward dynamic networks like Facebook, Instagram, or LinkedIn. These networks are more complex, as they constantly change

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and are often modeled as temporal graphs, where the relationships among people shift progressively. It is difficult to find important nodes on OSM, and this topic is receiving more research attention [8]. OSM platforms are often more effective communication channels than phone calls or face-to-face meetings. Nowadays, researchers are increasingly using social network-based modeling in many areas, such as fraud detection, quotation analysis, and marketing [9]. In social networks, the actions of one node can affect its connected nodes and spread to others through specific interaction rules. This spreading process is called diffusion [10]. Diffusion is important in network analysis because it helps to explain how information, behavior, or influence spreads through a network. For example, in educational networks, diffusion can show how knowledge moves between students and teachers [11]. In healthcare, such models can be used to simulate the spread of diseases among patients [12]. In marketing, it reflects how promotional messages or product information reach potential customers [13] and [14]. Graph Neural Networks (GNNs) [15], [16] have achieved noteworthy success in learning from graph-structured data by effectively leveraging node features with local neighborhood information [17]. However, classical GNNs remain limited in their ability to capture global or hierarchical patterns. Particularly, those involved in long-range dependencies or tree-like structures within the graph [18]. This limitation prevents GNNs from fully utilizing the social network's underlying structural wealth. To address this limitation of GNNs, a novel method using the Prufer sequence [19] has been proposed in this paper. The Prufer sequence is a succinct representation of a spanning tree that introduces richer topological information into learning. It constructs a Minimum Spanning Tree (MST) of the Facebook social network graph and obtains the associated Prufer sequence. A structural feature set is then derived from this Prufer sequence, including label frequency, entropy, transition influence, and subtree depth, which provides information about node roles and their positions within the network hierarchy. These features are then integrated with standard centrality measures and input into a GraphSAGE model [20], [21] for multi-class node classification by structural depth.

GraphSAGE has been used in this research, as it allows inductive learning and the sampling and aggregation of neighborhood features, which makes it well-suited for large networks. To support both conventional and deep learning models, a variety of graph structural features have been explored together. Node importance is captured by metrics like PageRank, degree centrality, and closeness centrality, which are based on patterns of local and global connectivity. In tasks like role identification, influence estimation, and graph embedding, these features have proven useful. These measures often focus on localized patterns or steady-state behaviors, while lacking the ability to capture the hierarchical structure or spanning substructure of graphs. This has motivated the study of tree-based representations, in which the Prufer sequence [19] serves as a compact encoding of a

labeled tree. Prufer sequences uniquely encode a spanning tree of an n -node graph into a sequence of $n - 2$ integers, preserving the structural hierarchy. They have traditionally been used in combinatorics and graph enumeration, but remain underutilized in modern graph learning pipelines. Although both GNNs and graph-theoretic encodings have been developed, only a few studies have attempted to integrate them. There has been little exploration of how global structural encodings can contribute to node features in GNN-based models, particularly high-order encodings derived from global structures such as Prufer sequences derived from MSTs. This paper addresses this gap and proposes the use of Prufer-derived features in a GraphSAGE architecture to leverage both local and tree-based global structure in a node classification setting. The main additions of the paper are as follows:

- Proposed an approach to social media node classification by integrating Prufer sequence-based structural features with GNNs.
- Introduced node-level Prufer sequence descriptors derived from the MST to encode global hierarchical structure within the graph.
- Designed multiple Prufer-based statistics, including occurrence frequency, positional encoding, transition influence, neighborhood aggregation, and local entropy.
- Integrated the proposed tree-based structural encodings into a GraphSAGE framework.
- A comprehensive feature construction strategy is developed by combining Prufer-based properties with conventional graph metrics.
- The proposed PruferNET framework achieves a test accuracy of 99.63%, compared to 96.54% with traditional features.
- Through systematic ablation experiments, we demonstrate that the proposed encodings provide complementary structural information beyond classical centrality-based features.

The main contribution of this work is not the development of a new GNN architecture, but the introduction of a tree-based global structural encoding that can be integrated with existing GNN models. In particular, we show that features derived from the Prufer sequence of an MST capture global hierarchical roles and connectivity patterns that are not readily obtained through local message-passing mechanisms alone. Unlike tree-based neural models that require the input graph to be a tree, our method does not assume that the original graph has a tree structure and does not replace the existing GNN model. Instead, we construct an MST only to extract Prufer-based structural features. These features are then combined with the original node attributes. In this way, the standard GNN architecture remains unchanged, while additional global structural information is incorporated into the learning process. To the best of our knowledge, this work is among the first to systematically explore Prufer-sequence-based node features as a form of global structural encoding for node classification in large-scale social networks. This

study focuses on finding important nodes in the Facebook social network. The study also examines how they help spread information, increase user activity, and shape the behavior of the network. The study applies network analysis methods to identify key individuals or groups that significantly influence how information spreads, user opinions and behavior. The primary objective is to identify the most influential individuals in a Facebook social network using both standard graph features and advanced topological features based on Prufer sequences.

The remaining parts of the paper are arranged as follows: Sections II, III, and IV present the research background, motivation, goals and objectives, research gaps, and main contributions. The detailed methodology and the analysis of experiments and results are presented in Sections V and VI, respectively. Section VII provides a summary of our work, followed by the references.

II. RESEARCH BACKGROUND

Social networks are continuously transforming the way people interact daily and significantly influence opinions and behaviors because of their widespread popularity [22]. Social influence occurs when people adjust their actions or beliefs based on others, driven by factors such as relationships and characteristics of the network [23]. This phenomenon affects various domains, such as healthcare [24], online recommendations [25], career decisions [26], rumor propagation [27], political outreach [28], and also in the spread of diseases [29]. Previous studies by [30] and [31] have examined social influence in these types of networks. A key objective in social network analysis is to identify influential users, as they play significant roles in controlling rumor propagation [32], [33], managing disease outbreaks [34], [35], and enhancing marketing strategies [36]. In recent years, researchers have shown a growing interest in studying how rumors spread in online social networks (OSNs). As a result, various methods have been developed to analyze this phenomenon [37]. The existing research can be generally grouped into two main classes: macroscopic and microscopic. Macroscopic approaches focus on observing the overall dynamics of rumor spreading. In contrast, microscopic approaches focus on individual interactions, such as identifying “who influenced whom”. To better understand word-of-mouth propagation in OSNs, several studies [38], [39] have adapted these models. The fundamental point of these research papers is the integration of topological features in the networks. Microscopic-level methods have gained significant attention because they enable detailed analysis of individual interactions. So, they have great effects on rumor spread within social networks [40], [41].

Recent research in social network analysis has focused mainly on two key areas: community detection and influential node identification [42]. Many existing methods try to pinpoint influential users, but often overlook or fail to recognize the predictive aspect of influence. For example, the study by Mousavi et al. [43] took into account both structural

similarities and user action similarities within social networks. They developed a detailed table of user actions to track activities across the network, which consequently improved community detection. Similarly, Wang et al. [44] proposed integration of network graph structures with user profile attributes to identify overlapping social communities. In another study, Hu and Yang [45] suggested a clustering edge-based community detection method of a social network graph. They measured structural similarity using the Jaccard index for edge pairs that shared a common node. It actually assigns zero similarity to pairs that did not match. In parallel, Vathi et al. [46] proposed similarity criteria based on structural aspects (e.g., following, common followers, friends), content aspects (e.g., hashtags), and interaction aspects (e.g., replies, retweets, mentions). They used the affinity propagation algorithm for community detection, which revealed hidden relationships and insights within the networks. Community detection is still a major challenging task in social network analysis. Particularly, the identification of similar node groups is necessary to understand their key role in understanding network structures. The approaches in [47] and [48] used optimization techniques to address influential users. Specifically, Zareie et al. [48] used the Grey Wolf Optimization (GWO) algorithm. Similarly, Mohammadi and Saraee [47] applied a multi-objective optimization approach to identify the most influential users through evolutionary search. Both methods are useful for evaluating various solutions to deliver the optimal set of influential nodes. Zhao et al. [49] analyzed node importance in various networks through centrality measures. Additionally, for community detection, research in [50] has used MST and modularity criteria. Prior studies have pointed out that recognition of influential nodes relying solely on metrics like follower count or centrality may not reflect their real influence [51]. Numerous factors affecting a user’s influence can vary with content and context. This suggests that user reaction metrics may serve as a more reliable basis for evaluating influence. Table 1 compares different methods on their community detection capabilities using parameters such as prediction (predict), trust, centrality, interactions (interact), and group identification. It shows the key differences and strengths among these parameters. Node classification in graphs has been widely studied using GNNs [52], [53]. GNNs are neural network architectures which are designed to work on non-Euclidean data. In this area, models such as GCNs [54], graph attention networks (GATs) [55]. Finally, GraphSAGE [21] is frequently used for the task of learning node representations by aggregating information from the local graph neighborhood. Prior research has incorporated tree-based or positional information into GNNs, including BFS-based encodings, shortest-path distance features, and hierarchical pooling strategies. Existing approaches to structural encoding in GNNs include Laplacian positional encodings, random walk features, centrality measures, and distance-based encodings. These methods generally emphasize local neighborhood structures or rely on

TABLE 1. A concise representation of methods for influential node identification and community detection.

Approach	Predict	Trust	Centrality	Interact	Group	Used Dataset
Community finding using modularity maximization [51]	X	X	✓	X	✓	Slashdot
Edge clustering via structural similarity [45]	X	X	✓	X	X	Email, Web, Reality, Brightkite
MST and modularity-based community discovery [50]	X	X	✓	X	✓	Karate Club, Football, Books
Label propagation with centrality metrics [49]	X	X	✓	X	✓	Football, Email, Facebook, PGP
Social circle detection using profile [44]	X	X	✓	X	✓	Email, Web, Reality, Brightkite
Similarity via structure, hashtags, and behavior [46]	X	X	✓	✓	✓	Twitter
Action pattern mining for community detection [43]	X	X	✓	X	✓	Karate Club, Dolphins, Football
Multi-objective optimization for user ranking [47]	X	X	✓	X	X	WikiVote, NetHEPT, Epinions
Influencer detection using GWO algorithm [48]	X	X	✓	X	X	Hamsterster, Pretty Good, Astro
Influence forecasting using local reaction data [42]	✓	✓	✓	✓	✓	Twitter-Dynamic-Net
PruferNET (The Proposed Method)	✓	✓	✓	✓	✓	Facebook (SNAP), Email-Eu-core, CiteSeer, PubMed, Cora, Computers, Photo, Physics, CS, WikiCS.

learned positional representations during training. In contrast, the proposed approach constructs a global MST of the graph and derives node-level structural descriptors from its corresponding Prufer sequence. This encoding process is deterministic, non-parametric, and independent of model training. Consequently, it introduces complementary global structural information that is not explicitly captured by conventional message-passing mechanisms in standard GNN architectures.

III. MOTIVATION FOR THE RESEARCH

In an increasingly digital world, OSNs (e.g., Facebook) have become a key means of shaping opinions, behaviors, and information flows. There are millions of users in these networks. A relatively small group of users, often referred to as influencers, disproportionately shape how ideas move across networks. Identifying these influential nodes is important for a variety of purposes, particularly for efficiently designing viral or marketing campaigns and for combating misinformation and increasing community involvement. Traditionally, users have relied on centrality-based metrics (such as degree or PageRank) to estimate influence in a network. However, these methods are limited in their ability to model the underlying structural and hierarchical relationships among users. In addition, social influence is not determined solely by connectivity; trust, reachability, and user position in the underlying communication tree also contribute.

This research proposes a different approach to modeling social influence by integrating complex structural features, including trust-based metrics from prior work and graph-theoretic properties derived from the Prufer sequence (a compact encoding of spanning trees that captures both influence paths and user connectivity patterns). These structural representations are integrated into a GraphSAGE architecture to learn stable node representations that effectively distinguish between deeply embedded (influential) and shallow (peripheral) users in the Facebook network. Ultimately, this hybrid approach—focusing on

trust, topology, and sequence-based structure—addresses the weaknesses of purely statistical or shallow methods and incorporates a promising approach for influencer identification in high-complexity social graphs. In general, the effectiveness of this hybrid proposal is validated by improved performance in node classification, which presents a real-world opportunity to combine classical graph-based models and cutting-edge GNNs.

IV. RESEARCH GAPS AND CONTRIBUTIONS

Despite extensive research on influencer identification in social networks, several important research gaps remain. The most notable gaps are outlined below.

- **Dependence on agent-based and dynamic models:** Many studies have focused on agent-based simulation frameworks or dynamic trust models, primarily designed for applications in marketing and socio-economic domains. In these cases, trust is often interpreted narrowly, for example, as checking for spam or verifying offers, without considering its role within the overall network structure.
- **Limited unified framework for node attributes:** The majority of current methods deal with individual elements of a node's impact, for example, degree, betweenness, or efficiency of content propagation. Fewer studies have applied an integrated set of indicators—merging structural properties, trust, and patterns of interaction—to provide a comprehensive identification of influential nodes.
- **Lack of adequate attention to predictive modeling of future leaders:** While techniques for determining current influential nodes are thoroughly examined, less research has focused on predicting future influential nodes. Existing methods primarily focus on link prediction or network change, but not necessarily on modeling the influence trajectory over time.
- **Scalability limitations of traditional methods:** Statistical and topological methods often fail to scale effectively with large heterogeneous data. In contrast, machine learning models often face challenges when

dealing with complex networks due to limited labeled data and low interpretability.

- **Underutilization of combinatorial graph-theoretic tools:** Prufer sequences can capture structural links and hidden hierarchies in trees. However, such tools have not been widely explored for influencer node detection or graph pattern learning. This limits the ability to spot more complex structural patterns beyond basic centrality measures.

This research addresses the following points:

- **Integration of hybrid features:** This new design integrates structural features such as degree, closeness, and PageRank with trust-based measures such as average neighbor PageRank. It also incorporates advanced graph-theoretic features derived from Prufer sequences. Those include label frequency, entropy, unique label ratio, subtree depth, and transition influence.
- **Classify GNN-based leader node:** With the help of these enriched features, a GraphSAGE model classifies nodes as Very Shallow, Shallow, or Deep. These categories effectively represent non-influential and influential roles in the network.
- **Tree-based control paths modeling:** The model implicitly learns from spanning influence paths, thus providing a new way to identify prospective future leaders by extracting MSTs from the Facebook graph and encoding their structures using Prufer sequences.
- **Validation using real-world Facebook data:** The method was applied and tested on the Facebook SNAP dataset. It is seen from the results that classification accuracy improves when Prufer sequence-based features are used. This also highlights their significant effect on representing hierarchical path influence.
- **Interpretable dominance score:** The combination of trust scores and tree-based structures produces influence scores to make it easier for interpretation than less transparent deep models.

The main contribution of this work is the integration of a tree-based global structural representation with standard GNN message passing. Unlike existing approaches that rely on local measures such as degree, centrality, or positional encodings, our method captures the global tree structure of the graph. We extract features from the MST using Prufer sequences to represent hierarchical relationships and node roles within the topology. These features are combined with GraphSAGE to enrich node representations. Rather than proposing a new GNN architecture, this work demonstrates that incorporating global structural features can significantly improve classification performance.

V. METHODOLOGY

This section describes the method, namely PruferNET, used to identify influential users in the Facebook social network. The represented method unifies classical graph-theoretic features, trust-based metrics, and Prufer sequence-based structural attributes within a GNN framework. Our methodology

comprises six main stages: data acquisition and preprocessing, feature extraction, MST generation, Prufer sequence computation, model training with GraphSAGE, and evaluation. The workflow of the proposed methodology is represented in Figure 1. Although the node labels are defined by subtree depth in the MST, no depth-related features are included as input. Specifically, subtree depth, distance from the root, and any direct transformation of these quantities are not used in the feature set. The proposed Prufer-sequence-based features are computed only from structural properties of the spanning tree, such as how often a node appears in the Prufer sequence, its position in the sequence, transition patterns, and local Prufer entropy. The structural role of a node without directly encoding its depth, thereby preventing label leakage.

The proposed PruferNET framework is summarized in Algorithm 1. It begins by extracting fundamental structural features such as centrality and PageRank, followed by the computation of a trust score that reflects the average importance of neighboring nodes. Edge weights are then assigned based on the node degrees. This enables the construction of an MST that captures the backbone structure of the graph. From this tree, a Prufer sequence is generated through an iterative leaf-removal process as described in Algorithm 2. Algorithm 2 serves as the basis for deriving additional features such as frequency, entropy, uniqueness, transition influence, and subtree depth. These Prufer-based features are then combined with structural and trust-based features to form an enriched representation. The learning process is carried out using a two-layer GraphSAGE model, as detailed in Algorithm 3. Here, the model is trained and validated iteratively to preserve the best-performing model parameters. Finally, predictions are generated and evaluated across different node-depth categories to assess the contribution of the Prufer features to the overall performance.

A. DATA SOURCE AND PREPROCESSING

Our research utilizes the Facebook Social Circles dataset, which is openly accessible through the Stanford Network Analysis Project (SNAP) ([56]). This dataset represents an undirected, unweighted graph in which each node represents a user, and each edge signifies a friendship between two users. The dataset is first loaded as a graph, and only the largest connected component is retained to ensure consistent structural analysis. Here, self-loops and duplicate edges are also removed.

B. ADDITIONAL DATASET: EMAIL-EU-CORE [57]

To further validate the generalization capability of the proposed PruferNET framework, we additionally evaluate the model on the Email-Eu-core dataset. The Email-Eu-core network dataset exhibits a different structural organization and connectivity pattern. This dataset represents email communication between members of a European research institution and contains 986 nodes and 16,064 edges.

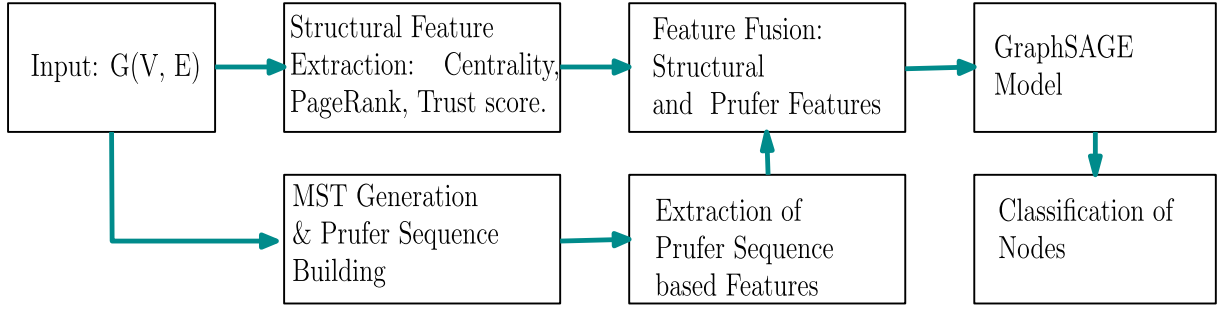


FIGURE 1. Proposed PruferNET workflow model.

Algorithm 1 Proposed PruferNET Framework

Input: Graph $G = (V, E)$
Output: Predicted labels with and without Prufer features

$F_s \leftarrow$ structural features (degree, closeness, PageRank);
 Compute trust score: $T(v) = \frac{1}{\max(1, \deg(v))} \sum_{u \in N(v)} p(u)$;

foreach $(u, v) \in E$ **do**
 $D_u \leftarrow \deg(u)$; $D_v \leftarrow \deg(v)$;
 if $(D_u + D_v) \neq 0$ **then**
 $w(u, v) \leftarrow \frac{1}{D_u + D_v}$;
 else
 $w(u, v) \leftarrow 0$;
 end
end

end
 Construct MST T from G using weights $w(u, v)$;
 Compute Prufer sequence P from T using Algorithm 2;
 Extract Prufer features F_p (frequency, entropy, uniqueness, edge transition influence, subtree depth);
 $\mathbf{x}_{with} \leftarrow F_s \cup T(v) \cup F_p$;
 $\mathbf{x}_{without} \leftarrow F_s \cup T(v)$;
 Normalize features;
 Compute node depth $\delta(v)$ in T and group nodes based on percentiles;
 Split dataset into train (60%), validation (20%), test (20%);
 $\hat{y}_{with} \leftarrow \text{TrainEval}(\mathbf{x}_{with})$;
 $\hat{y}_{without} \leftarrow \text{TrainEval}(\mathbf{x}_{without})$;
return $\hat{y}_{with}, \hat{y}_{without}$;

The task remains to classify nodes into three depth-based structural categories: Very Shallow, Shallow, and Deep. This additional evaluation allows us to assess the robustness and generalization capability of the proposed Prufer-based structural encoding across networks with varying topological characteristics.

C. STRUCTURAL FEATURE EXTRACTION

To characterize the position and strength of effect for each node inside the network, the following centrality-based features are primarily calculated:

Algorithm 2 Prufer Sequence Construction From the MST

Input: Tree $T = (V_T, E_T)$
Output: Prufer sequence P

$P \leftarrow \emptyset$;
 $n \leftarrow |V_T|$;
while $n > 2$ **do**
 Select the leaf vertex $v \in V_T$ with the smallest label;
 Let u be the unique neighbor of v ;
 Append $\text{label}(u)$ to P ;
 Remove v and the edge (u, v) from T ;
 $n \leftarrow n - 1$;
end
return Prufer sequence P ;

- 1) **Degree centrality** [58], [59]: The degree centrality evaluates the number of explicit links a node possesses within a network. In such a case of a directed graph, this metric provides insight into how immediately a node may be influenced by any entity. These can be information or activity, propagating through the system. The degree centrality $C_D(v)$ of a node v is mathematically defined as:

$$C_D(v) = \frac{\deg(v)}{n - 1} \quad (1)$$

where:

- $\deg(v)$ depicts the total sum of edges incident to the node v ,
 - n denotes the entire number of nodes present in the graph.
- 2) **Closeness centrality** [60], [61]: Closeness centrality defines the measurement of how near a node is to all the other nodes in a network, depending on the shortest path distances. It reflects the efficiency with which a node can spread information or interact with others in the system. This value is obtained by taking the reciprocal of the sum of the shortest path lengths from a given node to all remaining nodes. A node with higher closeness centrality can reach other nodes more quickly across the network. The closeness centrality $C_C(v)$ of a

Algorithm 3 Training and Evaluation Procedure for GraphSAGE

Input: Feature matrix $\mathbf{X}$, graph $G = (V, E)$, label vector $\mathbf{y}$, and train/validation/test splits

Output: Predicted labels $\hat{\mathbf{y}}$ and performance metrics

Initialize model parameters $\mathbf{W}$;

Assign node features $\mathbf{X}$ and labels $\mathbf{y}$ to the model;

Configure a two-layer GraphSAGE network with dimensions ($d_{\text{in}} \rightarrow 64 \rightarrow C$) and dropout rate 0.5;

Set optimizer as AdamW with learning rate $\eta = 0.003$ and weight decay $\lambda = 10^{-4}$;

Define the loss function as negative log-likelihood loss $\mathcal{L}_{\text{NLL}}$;

Initialize best validation accuracy $A_{\text{val}}^* \leftarrow 0$ and optimal parameters $\mathbf{W}^* \leftarrow \emptyset$;

for $\text{epoch} = 1$ **to** 300 **do**

Perform forward propagation on training nodes using graph G ;

Compute training loss $\mathcal{L}_{\text{train}}$;

Perform backpropagation and update parameters $\mathbf{W}$;

Evaluate model on validation set to obtain accuracy A_{val} ;

if $A_{\text{val}} > A_{\text{val}}^*$ **then**

$A_{\text{val}}^* \leftarrow A_{\text{val}}$;

Save optimal parameters $\mathbf{W}^* \leftarrow \mathbf{W}$;

end

end

Load optimal parameters $\mathbf{W}^*$ into the model;

Generate predictions $\hat{\mathbf{y}}$ on the test set;

Compute performance metrics (e.g., Accuracy, Precision, Recall, F1-score) across node-depth levels: {Very Shallow, Shallow, Deep};

return $\hat{\mathbf{y}}$ and performance metrics;

node v is expressed as:

$$C_C(v) = \frac{n-1}{\sum_{u \neq v} d(v, u)} \quad (2)$$

where:

- n denotes the total number of nodes in the network,
- $d(v, u)$ represents the length of the shortest path that exists between node v and node u .

- 3) **PageRank score** [62], [63]: PageRank centrality evaluates the significance of a node by considering the importance of its neighboring nodes rather than merely counting direct connections. The algorithm yields a probability distribution that shows how likely each node in the network is to be visited. The PageRank $PR(v)$ of a node v value can be formulated as:

$$PR(v) = \frac{1-d}{n} + d \sum_{u \in \text{In}(v)} \frac{PR(u)}{\deg^+(u)} \quad (3)$$

where:

- d is the reduction factor, often set to a value of 0.85,
 - n calculates the total number of nodes count in the network,
 - $\text{In}(v)$ denotes the set of nodes linked to v ,
 - $\deg^+(u)$ shows the outgoing edges from node u ,
 - $PR(u)$ is equal to the PageRank metric for node u .
- 4) **Trust Score** [64]: In addition, the trust score is also introduced to measure a node's reliance based on its neighboring nodes. The metric represents the mean PageRank of connected nodes, which indicates a node's local authority and influence. The trust score $T(v)$ is defined as shown below in Equation 4.

$$T(v) = \frac{1}{d(v)} \sum_{u \in N(v)} PR(u) \quad (4)$$

where:

- $d(v)$ shows the degree of node v ,
- $N(v)$ is related to the set of neighbor nodes connected to v ,
- $PR(u)$ is for the associated PageRank score of node u .

D. MST CONSTRUCTION AND PRUFER SEQUENCE EVALUATION

An MST is created for seeing the in-depth structural connectivity, which uses Equation 6. This MST shows the main connections in that network, making tree-based analysis easier.

$$T^* = \arg \min_{T \subseteq G} \sum_{(u,v) \in T} w(u, v) \quad (5)$$

subject to T being a spanning tree of G

where:

- T^* represents the MST of the graph G ,
- $w(u, v)$ denotes the weight associated with the edge connecting vertices u and v ,
- the minimization is performed over all possible spanning trees $T \subseteq G$.

The edge weight between two nodes is determined using their assigned degrees, as defined below:

$$\text{Weight}(u, v) = \begin{cases} \frac{1}{\deg(u) + \deg(v)}, & \text{if } \deg(u) + \deg(v) \neq 0, \\ 0, & \text{otherwise.} \end{cases} \quad (6)$$

Here, $\deg(u)$ and $\deg(v)$ represent the degree values of the nodes u and v , respectively. This formulation ensures that the sum of the node degrees inversely influences the edge weights, allowing nodes with higher combined degree values to contribute more to the overall tree cost. This formulation ensures the edge weight is inversely proportional to the combined degree of the two connected nodes. Edges that link highly connected nodes receive smaller weights. It leads the MST to favor less connected but structurally significant links.

From that MST, a unique encoding of the tree that captures its topology is calculated, known as the Prufer sequence. After that, several high-level features from the Prufer sequence are derived:

1) LABEL FREQUENCY

This measures how many times each node appears in the Prufer sequence derived from the MST [65]. Since this encoding sequence includes only internal nodes, a higher frequency means the node is more central in the tree. Those types of nodes act as the link or hub among subtrees. Suppose $P = (p_1, p_2, \dots, p_{n-2})$ be the Prufer sequence which corresponds to an MST of n nodes. The frequency $f(v)$ of a vertex $v \in V$ that appears in the sequence P can be defined as:

$$f(v) = \sum_{i=1}^{n-2} \mathbf{1}_{\{p_i=v\}} \quad (7)$$

In this expression, $\mathbf{1}_{\{p_i=v\}}$ is used as an indicator function. It takes the value 1 when $p_i = v$ and 0 otherwise. The term $f(v)$ tells the number how many times a node appears in the Prufer sequence. A higher value of $f(v)$ means that the corresponding node contributes more important connectivity to the spanning tree.

2) ENTROPY

Entropy measures the diversity of nodes in the distribution of their occurrence within the Prufer sequence. A higher entropy value indicates that multiple nodes participate more uniformly in the overall structure of the MST. A lower entropy value shows that only a small number of nodes appear frequently in the sequence. This is a clear indication of being dominant over these nodes to connect different parts of the MST. The entropy calculation follows the standard Shannon entropy formulation applied to the frequency distribution of the label [66], [67]. The mathematical expression of entropy is:

$$H(P) = - \sum_{v \in V} \frac{f(v)}{|P|} \log_2 \left(\frac{f(v)}{|P|} \right) \quad (8)$$

- P is the Prufer sequence which corresponds to the MST,
- $f(v)$ is the number of times node v arises in P ,
- $|P|$ is the total number of elements in the Prufer sequence,
- $H(P)$ shows how evenly the different nodes are distributed within the encoded sequence.

3) UNIQUENESS RATIO

This metric is the proportion of unique node labels that occur in the Prufer sequence relative to the total number of nodes in the graph. This metric is utilized to inform how many different nodes take part in the network and how equally the connections are spread across them. A high value of it implies that a larger number of nodes contribute to the formation of the MST structure. It is significant to have a

more balanced and decentralized network [67]. The definition of the uniqueness ratio is as follows:

$$U(P) = \frac{|\{v \mid f(v) > 0\}|}{|V|} \quad (9)$$

where:

- $f(v)$ is the number of times node v occurs in the Prufer sequence P ,
- $\{v \mid f(v) > 0\}$ is the set of nodes that occur at least once in P ,
- $|V|$ is the total number of nodes in the original graph, and
- $U(P)$ shows the share of nodes included in the Prufer encoding. This describes how extensively participation is spread across the network.

4) EDGE TRANSITION INFLUENCE

This feature is responsible for capturing the specific node-to-node transitions in the Prufer sequence. Next to each other, pairs of nodes are examined to understand how the influence spreads locally and how strongly different parts of the network are connected. When a node occurs often in these transitions, it refers to playing an important role in connecting different parts of the network. In structural graph encoding, the Edge Transition Influence (ETI) is introduced based on transition-based frequency analysis [68], [69].

Let the Prufer sequence corresponding to a tree with n vertices be represented as:

$$P = [p_1, p_2, \dots, p_{n-2}]$$

The set of consecutive node transitions within this sequence is expressed as:

$$T = \{(p_1, p_2), (p_2, p_3), \dots, (p_{n-3}, p_{n-2})\}$$

ETI of a vertex v is then defined as the total number of occurrences of v within these consecutive transitions:

$$I(v) = \sum_{u \in N(v)} (\text{count}((v, u)) + \text{count}((u, v))) \quad (10)$$

where:

- $N(v)$ indicates the set of vertices adjacent to v in the original graph,
- $\text{count}((v, u))$ represents the number of times the ordered pair (v, u) occurs consecutively in P ,
- $I(v)$ quantifies the level of influence or involvement of node v based on its consecutive appearances in the Prufer sequence.

5) SUBTREE DEPTH

This feature represents how far a node is from the designated root node within the MST. This shows how far a node (such as a person or account) is from the main 'root' of the structure of a network [70]. It is a way to measure influence or rank. Nodes near the root are usually big players or early spreaders of ideas, while nodes farther away are often less central or

newer followers in the network. Let the MST be represented as $T = (V, E_T)$, where V denotes the collection of vertices and E_T represents a collection of edges included in the tree. Let $r \in V$ be the designated root vertex, typically chosen as the most central or influential node. The depth of the subtree corresponding to a vertex $v \in V$ is expressed as:

$$\text{Depth}(v) = \text{dist}_T(r, v) \quad (11)$$

where $\text{dist}_T(r, v)$ indicates the shortest path distance from the root r to the vertex v within the tree T .

E. TREE DEPTH REFLECTS STRUCTURAL INFLUENCE

The depth of a node in the Prufer-derived MST provides a hierarchical view of its structural position within the network. In a spanning tree representation, nodes closer to the root tend to serve as connectors supporting multiple downstream branches, whereas deeper nodes typically reside toward the periphery of the structure. From a structural perspective, influential nodes often occupy positions that bridge multiple substructures or maintain connectivity across communities. Such nodes tend to appear higher in the tree hierarchy and contribute more frequently to the Prufer sequence. Conversely, nodes located deeper in the tree generally correspond to peripheral or less central participants in the network. Therefore, subtree depth serves as a proxy for hierarchical influence within the global topology. By learning from depth-based categories, the model captures distinctions between structurally central and peripheral nodes, which is consistent with classical graph-theoretic notions of centrality and influence.

F. FEATURE MATRIX CONSTRUCTION

To verify the suggested method's effectiveness, traditional and extended feature sets are used in this study. So, two distinct feature matrices were constructed. The first matrix is composed exclusively of traditional graph-theoretic features, including node degree centrality, closeness centrality, PageRank, and trust score, each capturing fundamental structural and influence-based properties of the graph. The second matrix extends this representation by incorporating additional structural features derived from the Prufer sequence. The Prufer-based features considered here are entropy, a measurement of the variability of node labels. Subtree depth indicates the hierarchical position of nodes within the reconstructed tree. At last, transition influence quantifies the effect of label changes in the Prufer sequence. All features in both matrices are normalized and standardized through a standard scaler. That is done to maintain consistency and improve the performance of the learning algorithm. Each feature is centered at zero and scaled to unit variance during this process. This preprocessing step is essential to avoid the dominance of large-magnitude features. It is consequently used to accelerate convergence in model training. This

procedure can be expressed as shown below in Equation 12.

$$\mathbf{X} = \begin{bmatrix} d_{\text{norm}}(v), C(v), PR(v), T(v), \\ f(v), H(P), U(P), I(v), \text{depth}(v) \end{bmatrix}. \quad (12)$$

Each row vector $\mathbf{X}_v$ corresponds to the feature representation of a node $v \in V$. The individual elements of this vector are described as follows:

- $d_{\text{norm}}(v)$: normalized degree of node v ,
- $C(v)$: clustering coefficient,
- $PR(v)$: PageRank score,
- $T(v)$: transition-based centrality value,
- $f(v)$: is the number of times node v occurs in the Prufer sequence P ,
- $H(P)$: entropy derived from the Prufer sequence P ,
- $U(P)$: uniqueness ratio obtained from P ,
- $I(v)$: influence index of node v ,
- $\text{depth}(v)$: level of node v within the spanning tree.

G. LABEL GENERATION

Meaningful node labels for classification are derived from the structural properties of the decoded Prufer tree. Each node $v \in V$ is given a category label according to its depth in the subtree structure. A node's depth shows how far it is from the root, or how close it is to the outer edge of the tree. The depth values of nodes are divided into three parts to group nodes into different structural categories based on rank-based thresholds. Let Q_1 and Q_2 denote the first and second quantiles (lower and upper medians) of the node depth distribution. Then the label assignment is defined as:

$$y(v) = \begin{cases} 0, & \text{if } \text{depth}(v) \leq Q_1, \\ 1, & \text{if } Q_1 < \text{depth}(v) \leq Q_2, \\ 2, & \text{if } \text{depth}(v) > Q_2. \end{cases} \quad (13)$$

This three-class labeling scheme captures the hierarchical positioning of nodes with finer granularity than binary partitioning. Shallow nodes represent peripheral or upper-layer elements, medium nodes correspond to intermediate structural roles, and deep nodes capture the most embedded positions within the tree. Such a scheme allows the model to learn richer distinctions among nodes with different hierarchical roles.

H. GRAPHSAGE MODEL ARCHITECTURE

In this study, the nodes are classified into two meaningful categories. In which a two-layer GraphSAGE model [71] is explicitly designed for node-level prediction tasks. GraphSAGE is particularly suitable for inductive learning on large graphs and aggregates features from sampled neighborhoods. The model employs a pair of graph convolutional layers based on the SAGEConv operation. Those enable each node to aggregate information from its neighbors. Each convolutional layer is followed by a ReLU (activation function) to incorporate non-linearity and enhance pattern learning. The first layer transforms the input features into a 64-dimensional hidden vector, and the subsequent layer transforms this

hidden representation to the target class-specific output space. Dropout (rate = 0.5) is introduced to mitigate overfitting, which randomly disables neurons in training. Model parameters are optimized using the AdamW optimizer, considering a learning rate of 0.003 and a weight decay of 10^{-4} , promoting stable convergence and providing regularization against over-parameterization. This configuration enables effective training of the model while allowing its performance to be evaluated on unseen data.

$$h_v^{(k)} = \sigma \left(W^{(k)} \cdot \text{CONCAT}(h_v^{(k-1)}, \text{AGG}\{h_u^{(k-1)} : u \in N(v)\}) \right) \quad (14)$$

where:

- $h_v^{(k)}$ indicates the embedding of node v at layer k ,
- AGG is an aggregate function (e.g., mean or sum) applied over the neighbors of v ,
- σ is an activation function (ReLU),
- $W^{(k)}$ is the learnable weight matrix at layer k .

$$\hat{y}_v = \text{softmax}(W \cdot h_v^{(K)}) \quad (15)$$

where:

- $\hat{y}_v$ is the predicted label distribution for node v ,
- $h_v^{(K)}$ represents the final embedding of node v after K layers,
- W is the learnable weight matrix used for classification.

Equations 14 and 15 formalize the GraphSAGE update mechanism and the computation of final predictions, respectively.

$$L = - \sum_{v \in V_{\text{train}}} y_v \cdot \log \hat{y}_v \quad (16)$$

where:

- V_{train} is the set of training nodes,
- y_v is the ground-truth label of node v ,
- $\hat{y}_v$ is the predicted probability distribution for node v .

Equation 16 defines the cross-entropy loss, which penalizes deviations between the true labels and the predicted probabilities over all nodes in the training set. The same training protocol, feature configuration, and hyperparameters used for the Facebook dataset were maintained for the Email-Eu-core dataset to ensure a fair comparison.

I. EVALUATION AND VISUALIZATION

Performance of the model is assessed considering accurately detecting metrics, including accuracy (Acc), precision (Pre), recall (Re), and F1-score (F1). These indicators provide a comprehensive view of the model's effectiveness in accurately detecting influential nodes within the network. A comparison is made between models using traditional features and models that include Prufer sequence-based structural features. This comparison emphasizes the significance of incorporating topological information into the learning process. Also, it underscores the role of topological information in the enhancement of classification accuracy and model performance. As shown in Figure 2, for clearer

interpretation, a subgraph with 2,000 nodes is extracted and visualized from the full Facebook network. In this visualization, influential nodes are clearly highlighted. The figure helps highlight where these nodes are located in the network and how they take part in overall structure formation. The positions, interconnections, and roles of nodes within the community network provide meaningful insights into the distribution of their influence. They are helpful to validate the model's predictions from a structural perspective.

J. OVERFITTING CONTROL AND GENERALIZATION ANALYSIS

To mitigate potential overfitting, several precautions were taken during training. A fixed train/validation/test split (60/20/20) was used with a reproducible random permutation to ensure consistent evaluation. Dropout regularization (rate = 0.5) and weight decay were applied during optimization. Model selection was performed based on validation accuracy, and the best-performing model was evaluated only on the held-out test set. To further examine generalization capability, we evaluated the proposed PruferNET framework on an additional dataset (Email-Eu-core) using the same experimental protocol and hyperparameters. The consistent performance improvement across both Facebook and Email-Eu-core networks suggests that the high accuracy is not due to dataset-specific overfitting. In addition to the fixed-split evaluation, we conducted 5-fold cross-validation on the Facebook dataset. The average accuracy remained stable across folds, indicating that the performance gains are consistent and not driven by a particular partition of the data.

VI. IMPLEMENTATION AND RESULTS ANALYSIS

The proposed model was implemented in Python with a machine configured with a T4 GPU and 16 GB RAM. The Facebook social network dataset was taken into consideration to assess the results of the suggested model. The dataset has been briefly discussed below:

A. FACEBOOK SOCIAL NETWORK DATASET

The Facebook social network dataset is a widely used standard for graph mining and social network analysis. The dataset was created by Stanford Network Analysis Project (SNAP) ([56]). After pre-processing the dataset, the graph consisted of 4,039 nodes and 88,234 edges ($|V|$) and edges ($|E|$) respectively. For centrality calculations, the graph was considered unweighted, while weights were applied during the construction of the MST. The purpose of the experiment is to categorize nodes as influential (Deep), moderately influential (Shallow), or non-influential (Very Shallow) based on their positions in the structure of the network. We trained the model for 300 epochs, and we picked the best validation accuracy to select the final model for testing. The following metric is used to evaluate model performance:

$$\text{Accuracy(Acc)} = \frac{TP + TN}{TP + TN + FP + FN} \quad (17)$$

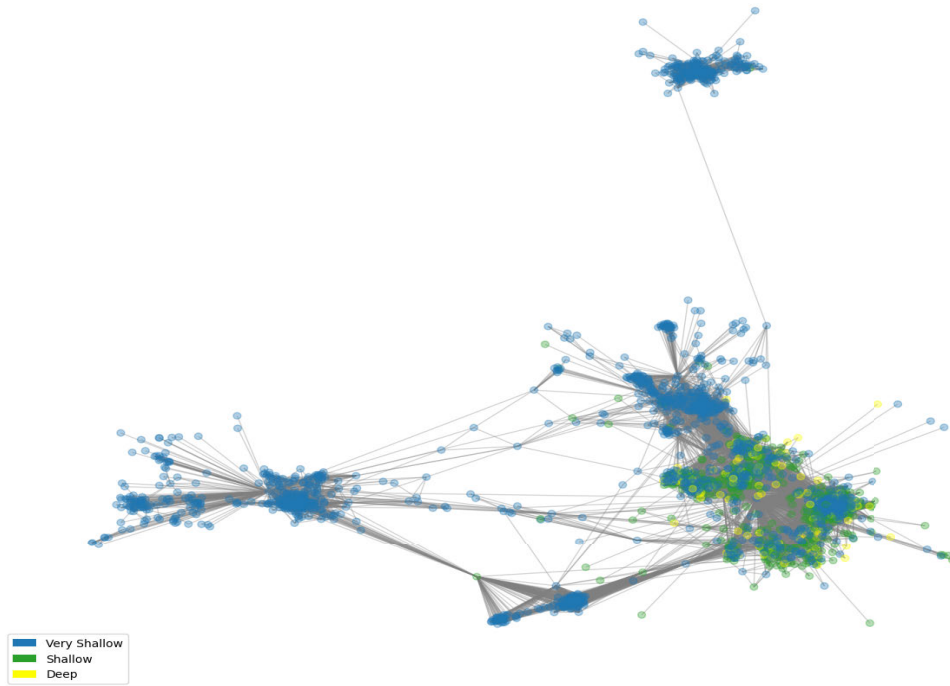


FIGURE 2. Detected influential users (yellow nodes) in the Facebook subgraph using Prufer-based features and GraphSAGE.

where, for a class C :

- **TP (True Positives):** Cases correctly predicted as belonging to the class C (for example, $C = \text{Deep}$, and Deep nodes are correctly classified as Deep).
- **TN (True Negatives):** Cases correctly predicted as not belonging to the class C . (for example, $C = \text{Deep}$, and the instances are neither Deep, nor are predicted as Deep).
- **FP (False Positives):** Cases incorrectly predicted to be in the class C (for example, $C = \text{Deep}$, Shallow or Very Shallow nodes wrongly predicted as Deep).
- **FN (False Negatives):** Cases incorrectly predicted as not in C (for example, $C = \text{Deep}$, Deep nodes wrongly predicted as Shallow or Very Shallow).

Apart from measuring temporal overlap, we also evaluated how well nodes were classified using a supervised learning setup. Standard classification metrics are utilized to obtain a comprehensive understanding of the model's behavior across the three structural categories (Figure 2). The model was trained with traditional structural graph features as well as the proposed Prufer sequence features, combined with a trust-based score.

The classification metrics were computed using the following standard definitions:

$$\text{Precision (Pre)} = \frac{TP}{TP + FP} \quad (18)$$

$$\text{Recall (Re)} = \frac{TP}{TP + FN} \quad (19)$$

$$\text{F1-Score (F1)} = \frac{2 \times \text{Pre} \times \text{Re}}{\text{Pre} + \text{Re}} \quad (20)$$

B. RESULTS ANALYSIS

The designed framework on the Facebook dataset was tested using classification experiments that included three target groups: Very Shallow, Shallow, and Deep. The performance of the constructed GraphSAGE model was evaluated under two experimental configurations: (i) including Prufer sequence-based features, and (ii) excluding them, serving as the baseline model. Table 2 presents the classification reports for both settings. The results of both experiments are discussed in the following section and are shown in Figure 4 and 5, respectively.

C. PERFORMANCE WITH PRUFER FEATURES ON FACEBOOK DATASET

The proposed model with Prufer features achieved a test accuracy of 99.63%, which shows an almost perfect classification in all three categories. The Very Shallow, Shallow, and Deep classes each achieved exceptionally high precision, recall, and F1-scores— 0.99, 1.00, and 1.00, respectively. These results indicate the outstanding reliability and consistency of this method. This is shown in Figure 4 for illustration. The results confirm that the addition of Prufer-based features and the trust score immensely improved the model's ability to distinguish between classes.

The 3D surface plot in Figure 4 visualizes the classification performance metrics— Accuracy, F1-score, Precision, and

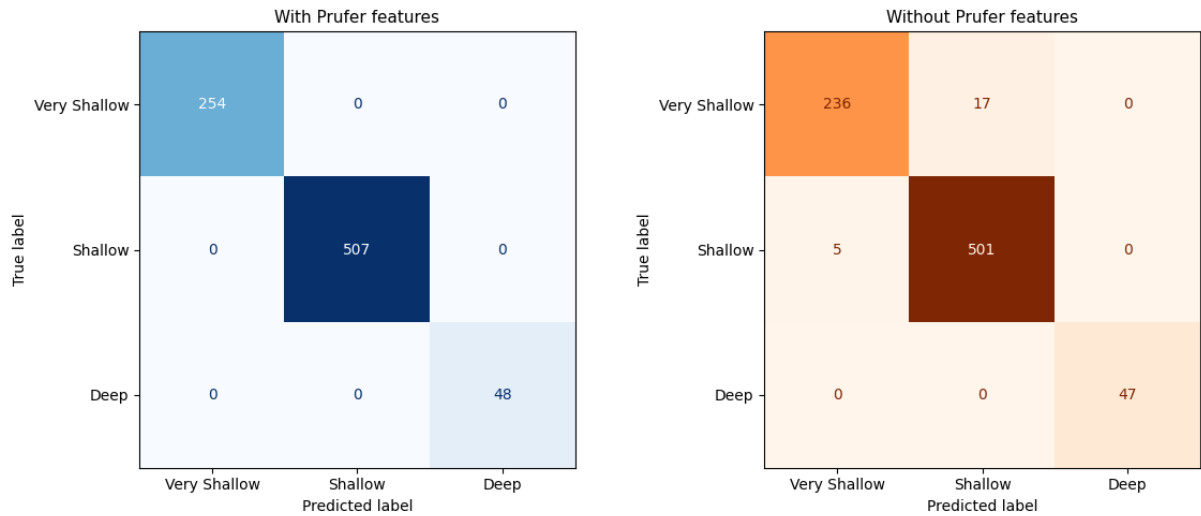


FIGURE 3. The confusion matrix of performance for both with Prufer features and without Prufer features on Facebook dataset.

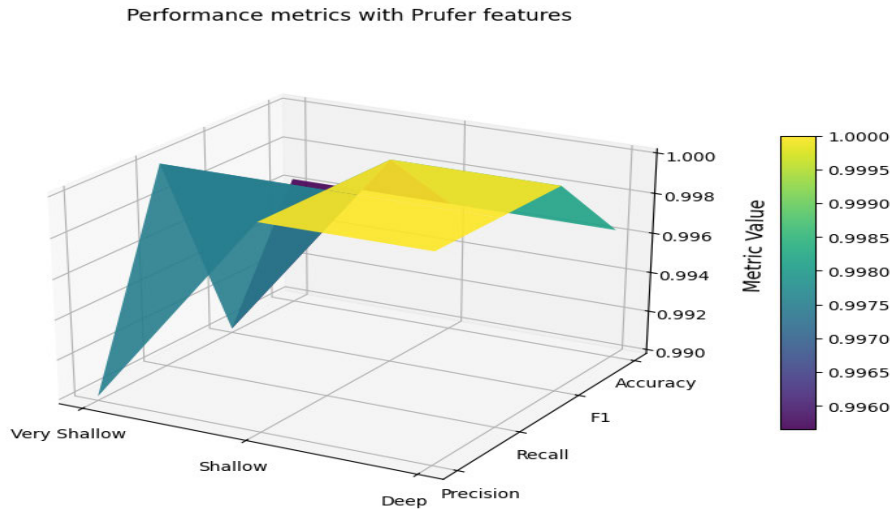


FIGURE 4. Results delivered by the model (using Prufer features) on Facebook dataset.

Recall. This plot represents the performance of the model using Prufer features and trust score across the three tree depth categories: Very Shallow, Shallow, and Deep. The X-axis represents the tree depth classes, the Y-axis indicates the metric type, and the Z-axis denotes the corresponding metric values. The color gradient from dark purple to bright yellow indicates enhanced performance, where both the surface height and color jointly represent the model's metric values. The bright yellow regions correspond to perfect scores (around 1.000), which means flawless classification for the Shallow and Deep classes. The slightly greener regions for the Very Shallow class represent marginally lower scores (around 0.99). This smooth color transition highlights the classifier's stability and consistency across classes. It is enough to indicate that Prufer sequence-based

features combined with the trust score improve both precision and reliability. The color contrast makes it easy to compare high and low-performing areas. It clearly shows the strengths and weaknesses of the model for inclusion in the research findings.

D. PERFORMANCE WITHOUT PRUFER FEATURES ON FACEBOOK DATASET

The basic GraphSAGE model reached a test accuracy of 96.54%. The results of the model without Prufer features are shown in Figure 5. Although the result's accuracy is good, this model has certain limitations. The Very Shallow class showed a weaker recall (0.93) value, which resulted in misclassifications into that class. Similarly, the Shallow class, although it achieved a strong recall (0.99) value,

it showed lower precision (0.94). Overall, the outcome indicated a higher rate of false positives. Table 2 presents the classification reports for both setups.

The 3D surface plot in Figure 5 illustrates the classification performance metrics. It shows the Precision, Recall, and F1-score of models trained without Prufer features but using the trust score, across three fundamental categories: Very Shallow, Shallow, and Deep. The X-axis represents the node depth categories, the Y-axis indicates the metric types, and the Z-axis shows the corresponding metric values. The color gradient from dark purple to light yellow indicates the increasing performance, where both the surface height and color intensity represent the metric value. The light yellow regions correspond to the highest performance values (around 0.98–0.99). It indicates the best classification performance, especially in the Deep category. The orange to light purple regions represent relatively high value, ranging from 0.96–0.97. It is typically used to correspond to the Shallow category. Whereas the dark purple areas indicate the lowest performance (around 0.93–0.95). This is mainly due to Recall in the Very Shallow category. The smooth transition across the surface suggests that the classifier is stable and shows minimal variation across different depth categories. This visualization clearly illustrates the trade-offs among Precision, Recall, and F1-score. This exhibits that the trust score maintains balanced performance across different node depth categories, with only slight decreases in the shallower categories.

E. GENERALIZATION ANALYSIS ON EMAIL-EU-CORE

To verify that the proposed method is not dataset-specific, we further evaluated PruferNET on the Email-Eu-core network. The classification performances with and without Prufer features are shown in Table 4. The baseline GraphSAGE model achieves a test accuracy of 0.8030 with a macro-F1 score of 0.79. After incorporating the Prufer-based structural features, the accuracy increases to 0.9343 and the macro-F1 score to 0.93. Performance improves across all three classes, particularly in the Shallow and Deep categories, indicating that the proposed structural encoding generalizes well across different social network structures.

Figure 7 shows the class-wise performance of the PruferNET in the Email-Eu-core dataset after adding Prufer-based structural features. Precision, recall, and F1-score are compared for the Very Shallow, Shallow, and Deep classes. The overall test accuracy is 93.43%. The model performs almost perfectly for the Very Shallow class (0.99 for all metrics). For the Shallow class, precision is 0.93, and recall is 0.89, yielding an F1-score of 0.91. In the Deep class, recall (0.92) is slightly higher than precision (0.87), which indicates that most Deep samples are correctly identified with a few false positives. Here, all scores remain above 0.87, which reflects a stable and balanced performance across all classes. Figure 6 shows the class-wise results of the model without Prufer-based features in that dataset. Precision, recall, and F1-score are reported for the Very Shallow, Shallow, and

Deep classes. The overall test accuracy is 80.30%. The model performs well for the Very Shallow class (F1-score 0.95), but the performance drops for the Shallow and Deep classes. In the Shallow class, recall falls to 0.67. That leads to a lower F1-score of 0.74. For the Deep class, precision decreases to 0.63, although recall is 0.77. Therefore, the scores are lower and less balanced across classes, indicating that the model struggles to capture structural differences without Prufer features accurately.

F. COMPARATIVE ANALYSIS

From the analysis of the results on the Facebook dataset, it has been observed that the PruferNET model with Prufer sequence-based features delivered an improved accuracy of nearly +3% (from 96.54% to 99.63%). This can be clearly seen in Table 3 and in Figure 3. This validates its robustness and generalizability. The improvements in Pre, Re, and F1 of the results also illustrate that the method reduces misclassification bias between categories. The key factor in this improvement is the graph-theoretic representation introduced by Prufer sequences. These features encode latent structural relationships that are often overlooked by conventional neighborhood aggregation in GNNs. In addition, the trust score further calibrates the predictions and improves the reliability. The most significant gain has been found to occur in the Very Shallow class, where the recall improved from 0.93 to 1.00. Prufer features provide a complementary signal to GraphSAGE, enriching the representation space with global graph connectivity. This method performs well on large datasets, making it useful for other graph classification problems. This approach is especially well-suited in situations where capturing detailed structural differences capture is essential. The results confirm the effectiveness of integrating Prufer sequences and the trust score with the GraphSAGE framework. The implemented framework demonstrates excellent performance, achieving near-perfect Pre, Re, and F1 across all three classes. This outperforms the baseline model and establishes the value of graph-theoretic features in deep learning-based graph analysis.

G. ABLATION STUDY

To assess the effectiveness of the proposed Prufer sequence-based features, an ablation study was conducted in two different experimental configurations:

- 1) Structural features with advanced Prufer features and trust score
- 2) Structural features without Prufer features and trust score

Both configurations used the same GraphSAGE model, training hyperparameters, and standard structural features such as degree, closeness centrality, PageRank, and neighborhood-based trust score. The only difference was the inclusion of five Prufer-derived features: label frequency, entropy, unique label ratio, transition influence, and subtree

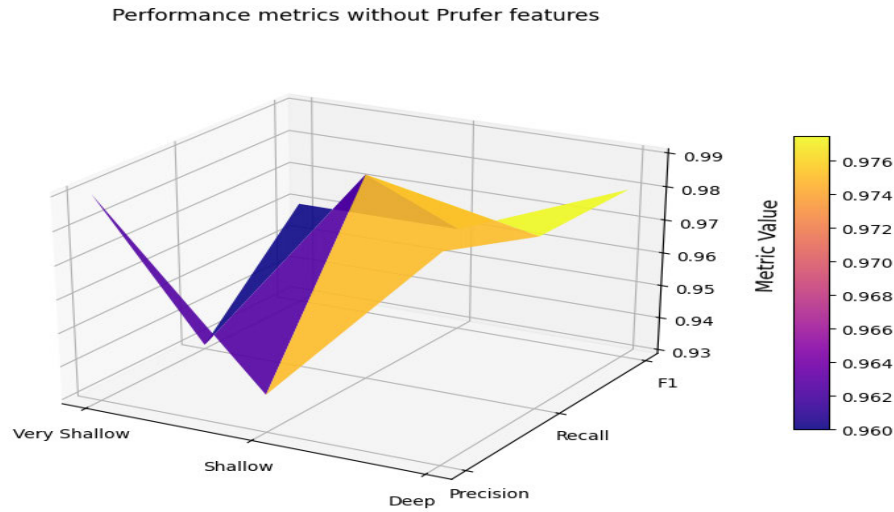


FIGURE 5. Results delivered by the model (without using Prufer features) on Facebook dataset.

TABLE 2. Classification performance on Facebook dataset with and without Prufer features.

Model	Class	Pre	Re	F1
Structural features with Prufer features (including trust score)	Very Shallow	0.99	1.00	0.99
	Shallow	1.00	1.00	1.00
	Deep	1.00	1.00	1.00
	Accuracy	0.9963		
Structural features without Prufer features (including trust score)	Very Shallow	0.99	0.93	0.96
	Shallow	0.94	0.99	0.96
	Deep	0.99	0.98	0.98
	Accuracy	0.9654		

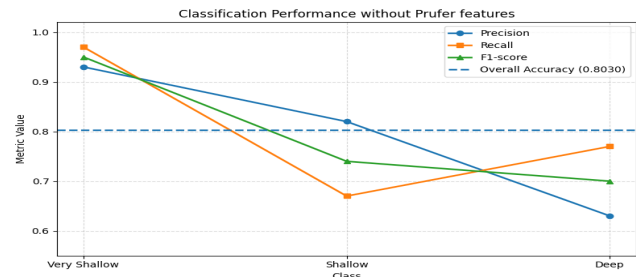


FIGURE 6. Performance of the model without Prufer-based features on Email-Eu-core dataset.

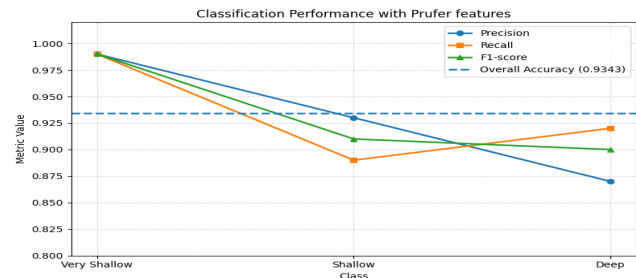


FIGURE 7. Performance of the model with Prufer-based features on Email-Eu-core dataset.

depth. To further eliminate possible circularity, we performed an ablation study in which all depth-related information was excluded from the input features. The model was trained

under two settings: (i) using only standard structural features such as degree, closeness centrality, PageRank, and trust score, and (ii) using the same features combined with Prufer-sequence-based encodings. In both cases, subtree depth was used only to define the class labels and was never included as an input feature. The consistent improvement observed after adding the Prufer-based features indicates that the performance gain is not due to implicit depth information, but rather to the additional global structural information captured by the Prufer representation. To further validate the contribution of the proposed structural encoding, we conduct ablation experiments by removing all Prufer-based features while keeping the GNN architecture and training protocol unchanged. The consistent performance improvement observed across both datasets demonstrates that the proposed encoding provides complementary structural information beyond conventional centrality-based features.

1) PERFORMANCE COMPARISON

The model that included Prufer-based features achieved a test accuracy of 99.63% on the Facebook dataset (based on Equation 17). As opposed to significantly outperforming the baseline model (with accuracy 96.54%) that only used traditional features and trust score. Both precision and recall showed noticeable improvements across the three classes, Very Shallow, Shallow, and Deep. This indicates better and more consistent identification of influential users. These

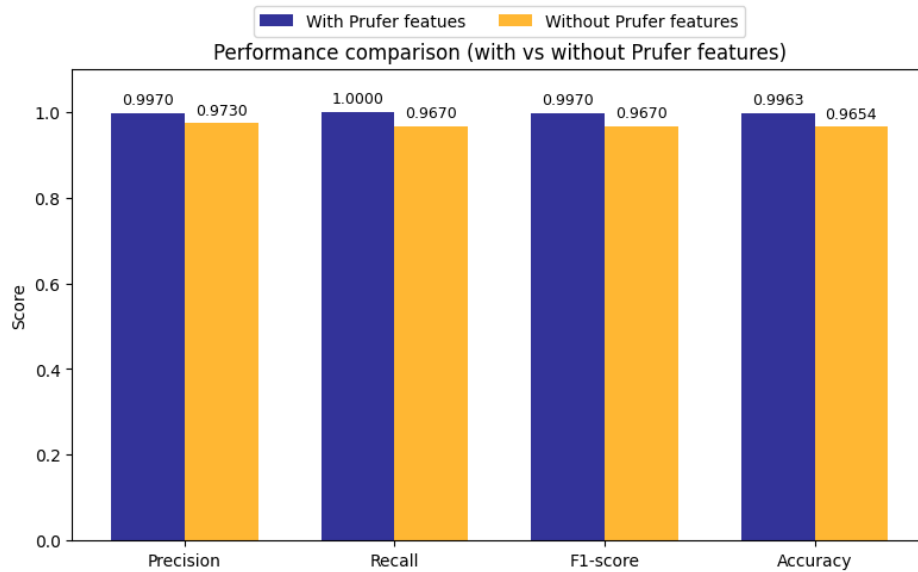


FIGURE 8. Performance comparison of all parameters with and without Prufer features on the Facebook dataset.

results are capable of showing that Prufer-based features help identify detailed patterns in the network's structure and hierarchy more effectively. These details are often missed by traditional centrality-based measures. Some features, like subtree depth and transitional influence, infer useful information about a node's actual position in the MST. These spatial details are closely linked to its influence on the social structure. This ablation study is shown in Figure 8. This bar chart clearly demonstrates that integration of structural features derived from the Prufer sequence prominently increases classification performance. These features enable the GNN to handle or model complex dependencies within the network more precisely. This leads to more reliable and robust detection of influential users.

The bar chart in Figure 8 gives an overall comparative graphical overview of macro-level performance measures, including Pre, Re, F1 and Acc. It reflects the Prufer feature-trained models and those without Prufer feature-trained models. Two contrasting colors are allotted for the differentiation of experimental settings: dark blue bars for the Prufer feature-trained model and orange bars for the no Prufer feature-trained model. Taller and denser blue bars are always about higher metric values (from 0.9963 to 1.0000), which imply better structural learning, better generalization, and better stability of classification accuracy. Shorter orange bars with a value of around 0.9654-0.9730 represent the performance decrease with the elimination of structural encoding as put forth by Prufer. The difference in bar size and saturation of colors unmistakably proves the strength contributed by Prufer features since the Pre, Re, and F1 measures for the model with them are all 0.99 and the accuracy is 0.9963, compared to being only 0.9654 without Prufer integration. This makes the comparison directly interpretable. Each color and bar height

uniquely characterizes the relative gain contributed by Prufer sequence-based structural representations. Overall, the figure indicates that introducing Prufer features markedly enhances discriminative power and reliability by a significant margin. It also affirms their positive effect on model stability and classification results in the Very Shallow, Shallow, and Deep classes.

To further evaluate the contribution of PruferNET, we compare the overall performance of the model under both experimental settings. The impact of incorporating Prufer-sequence-based structural features is clearly evident on the Email-Eu-core dataset also. When PruferNET is equipped with the proposed tree-based encoding, the test accuracy improves from 0.8030 to 0.9343 (shown in Table 4). There is also a clear improvement in the macro-F1 score, which increases from 0.79 to 0.93. These results have clarified that when we added global structural features from Prufer sequences, the model becomes much better at identifying the influential nodes.

H. COMPARISON WITH RECENT GRAPH-BASED NODE CLASSIFICATION METHODS

We have carried out a detailed evaluation to examine how effective, stable, and generalizable the proposed PruferNET framework is. For this purpose, we have performed a detailed comparative analysis based on three complementary perspectives: (i) A performance comparison on the Facebook influencer detection task is carried out, (ii) An evaluation across multiple real-world graph datasets was conducted, and (iii) A comparative analysis of the architecture against recent structure-enhanced GNN models introduced between 2023 and 2025 is done. This multi-level evaluation allows us to better understand the model. How the model improves

TABLE 3. Performance comparison on Facebook dataset with and without Prufer-based features.

Configuration	Acc	Pre (Macro)	Re (Macro)	F1 (Macro)
Structural features with Prufer features (including trust score)	0.9963	0.997	1.000	0.997
Structural features without Prufer features (including trust score)	0.9654	0.973	0.967	0.967

TABLE 4. Classification performance on Email-Eu-core dataset with and without Prufer features.

Model	Class	Pre	Re	F1
Structural features with Prufer features (including trust score)	Very Shallow	0.99	0.99	0.99
	Shallow	0.93	0.89	0.91
	Deep	0.87	0.92	0.90
	Accuracy	0.9343		
Structural features without Prufer features (including trust score)	Very Shallow	0.93	0.97	0.95
	Shallow	0.82	0.67	0.74
	Deep	0.63	0.77	0.70
	Accuracy	0.8030		

accuracy, how effectively it captures structural information, and how it performs in terms of computational cost and scalability in practice can be easily evaluated.

1) PERFORMANCE COMPARISON ON FACEBOOK INFLUENCER DETECTION

First, we compared PruferNET with several recent and influential node detection and classification methods on the Facebook social network dataset. In contrast to several existing methods that overlook explicit structural modeling, our framework is built to make full use of the network topology. It combines structural centrality measures such as degree, closeness, and PageRank with trust scores computed via neighbourhood-based and PageRank-based aggregation. In addition, it incorporates a global tree-based structural representation derived from Prufer sequences to capture the overall connectivity patterns of the network more effectively. The complete PruferNET model achieves a test accuracy of 99.63%, while the version without the Prufer-based features reaches 96.54%. This nearly 3% absolute increase in performance clearly shows in Table 5 that incorporating global spanning-tree information through Prufer encoding plays a significant role in the improvement of influential node classification.

Compared with DeepInfNode and GBDT-CGBO, which improve their results mainly through changes in model architecture or optimization techniques, PruferNET attains comparable or even better accuracy. PruferNET also introduces a lightweight structural enhancement based on graph topology. Unlike FG-SMOTE, which focuses on fairness through sampling, PruferNET is designed differently. Our approach focuses directly on identifying structural roles within the network and accurately detecting influential nodes based on topological characteristics.

2) CROSS-DATASET GENERALIZATION ANALYSIS

To further validate robustness beyond a single social network dataset, PruferNET was evaluated on multiple benchmark graphs spanning diverse domains, including citation networks

(Cora, CiteSeer, PubMed), co-purchase networks (Amazon Computers, Photo), co-authorship networks (Physics, CS), communication networks (Email-Eu-core), and hyperlink networks (WikiCS). Table 6 presents these results in detail.

Our results confirm that the model maintains robust performance across structurally heterogeneous graphs. In dense co-purchase and co-authorship networks, the proposed model achieves accuracies exceeding 94%. This demonstrates that the global spanning-tree encoding effectively captures structural roles. On sparse citation networks, the proposed model maintains competitive performance despite irregular connectivity patterns. These results indicate that the proposed tree-based structural abstraction generates transferable and domain-independent structural descriptors.

3) COMPARISON WITH RECENT GNN-BASED MODELS (2023–2025)

Recent research (shown in Table 7) has increasingly focused on the improvement of structural awareness in GNNs. It is mainly to overcome the inherent limitations of traditional message-passing frameworks, including GCN and GraphSAGE. Notable approaches include centrality-enhanced GCN variants, structural similarity aggregation networks, and transformer-based models. They also include structural positional encoding, frequency-aware harmonic encoders, and topology-regularized multi-branch GNN architectures.

Many recent methods try to improve performance by making the model deeper or by using complex spectral or transformer-based techniques, which require more computation. In contrast, PruferNET follows a different and simpler approach. Instead of changing how messages are passed between nodes, PruferNET improves each node by adding simple global structural information obtained from Prufer sequences. These features, taken from spanning trees, give an overall view of the graph's structure. They support the local neighborhood information while keeping the model simple and not making it much more complex. The proposed PruferNET consistently achieves better results than attention-based GNN models such as the GTAT family

TABLE 5. Performance comparison on Facebook influencer detection.

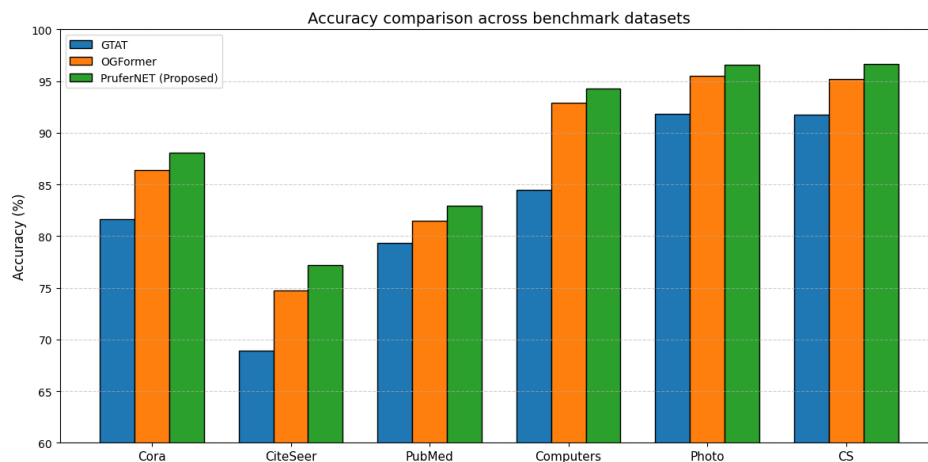
Method	Year	Test Accuracy	Structural Strategy
Proposed PruferNET (Full Model)	2026	99.63%	Structural features including trust score and global Prufer encoding
Proposed (Without Prufer features)	2026	96.54%	Structural features including trust score
DeepInfNode [72]	2023	99.10%	Centrality-enhanced GCN and epidemic modeling
GBDT-CGBO [73]	2023	98.76%	Gradient boosted decision and chaotic optimizer
FG-SMOTE [74]	2025	74.40%	Structural diversity and fairness-aware oversampling

TABLE 6. Performance of proposed PruferNET on multiple benchmark datasets.

Dataset	Accuracy (%)	Domain
CiteSeer [75]	77.20	Citation network
PubMed [76]	82.90	Biomedical citation network
Cora [77]	88.10	Citation network
Computers [78]	94.25	Amazon co-purchase network
Photo [79]	96.61	Amazon co-purchase network
Physics [80]	95.59	Co-authorship network
CS [81]	94.65	Co-authorship network
WikiCS [82]	96.00	Wikipedia hyperlink network

TABLE 7. Comparison with recent structure-enhanced GNN methods (2023–2025).

Method	Year	Structural Strategy	Model Complexity
DeepInfNode [72]	2023	Centrality-enhanced GCN with Susceptible–Infected–Recovered (SIR)	Moderate
IS-GNN [83]	2024	Influential and structural similarity-based node aggregation	High (similarity computation overhead)
GraphViz2Vec [84]	2024	Pre-embedding structure-aware encoding	Additional preprocessing stage
FG-SMOTE [74]	2025	Structural diversity with fairness-aware sampling	Oversampling-based training
OGFormer [85]	2025	Optimized attention-based graph transformer	High (attention-heavy architecture)
Harmonic Encoding GNN [86]	2025	Multi-scale harmonic structural encoding	Spectral projection overhead
GTAT [87]	2025	Cross-attention topology-aware modeling	High (attention-based architecture)
Proposed PruferNET	2026	Global spanning-tree encoding via Prufer sequence + trust score	Low–Moderate (lightweight augmentation)

**FIGURE 9.** Accuracy comparison of the proposed PruferNET with GTAT and OGFormer across benchmark datasets.

and recent graph Transformer models like OGFormer. Compared to GTAT, PruferNET improves the results by

about 3% to more than 20% on all benchmark datasets, with especially significant improvements on WikiCS and

CiteSeer. In comparison with OGFormer, which is a recent state-of-the-art model, our method achieves higher accuracy on five out of six common datasets. This shows that the addition of Prufer-based structural features is very effective. These results confirm that the proposed topology-aware encoding significantly enhances global structural learning while maintaining strong local consistency. Figure 9 presents how PruferNET performs in terms of accuracy compared to other recent graph attention and transformer-based models.

The extended comparison reflects that the model gives stable and reliable results on different types of networks, such as citation, co-purchase, co-authorship, and communication graphs. It helps the model understand the graph structure better without making the design more complicated or adding heavy attention or spectral layers. At the same time, the Prufer-based features are easy to understand from a graph theory point of view. Also, they are calculated only once before training, which makes the model suitable for larger graphs. In simple terms, PruferNET connects traditional spanning tree theory with modern GNNs. Thus, it manages to stay accurate while keeping the model easy to understand and efficient. This makes it suitable for large-scale social network and graph analysis tasks.

VII. CONCLUSION

A novel methodology to identify the influential users in large social networks is introduced in this paper. This method integrates sophisticated features extracted from Prufer sequences with a GNN architecture. Our empirical results indicate that incorporating Prufer-derived structural features, e.g., subtree depth, transition influence, and label frequency, significantly improves the accuracy achieved by the model in detecting influential nodes relative to traditional graph statistics alone. The ablation study validates the essentiality of these prominent features, as the model improves significantly in accuracy from 96.54% to 99.63%. It also balanced classification performance as well with the inclusion of Prufer-based features. Our results corroborate that the Prufer sequence can successfully capture the hierarchical and topological depth of this type of network. Most of the time, it is seen that conventional standard metrics generally have a tendency to ignore these types of subtle details of network structure. At the end, the result highlights the efficacy of Prufer sequence-based features as a strong dominator. There is a plan to find how subsequent studies could investigate how this method performs across a range of network types and changes of features based on Prufer sequences. It will calculate the overall improvement in diverse networks to find influential users and analyze networks.

In the future, we aim to broaden the scope of this methodology to dynamic or temporal graphs, where the network structure develops over time. Also, our plan is to adapt this approach for multi-class node classification in another dataset. That will be helpful to further validate the robustness and scalability of our technique, whether

integrating Prufer sequence representations within modern GNN architectures is fruitful or not.

DECLARATION

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